

BONUS points – report to PPG

- Create a PowerPoint presentation highlighting the major results, findings, and conclusions from your work.
- **You may earn up to 10 BONUS points.**
- Background/motivation material is not needed for the presentation.
- Presentation is open ended, but some recommendations to include:
 - Interesting visualizations from the EDA portion of the project.
 - Visualizations comparing model performance and selection of the best model.
 - Visualizations showing variable importances and/or coefficient summaries.
 - Visualizations showing model predictions with text interpreting the trends.

BONUS points – synthetic data

- In lecture, we have discussed the importance and usefulness of synthetic (fake) data in a complete Bayesian model workflow.
- **You may earn up to 15 BONUS points if you create your own synthetic data and demonstrate the ability to recover the model parameters that generated the data.**
- You may consider the regression problem OR the classification problem, but you must use Bayesian modeling techniques.
- Regardless you must use the following to earn the maximum bonus:
 - You must consider 1 categorical variable with 4 levels (unique values)
 - You must consider 3 continuous variables
 - You must specify the true functional (basis) relationship between the linear predictor and the inputs. You must specify the true parameter values.
 - You must generate small, medium, and large sample size data sets.
 - You must fit your model, assuming the correct functional (basis) relationship for the small, medium, and large sample sizes.
 - How well are you able to recover the true parameter values given the three training sample sizes?

BONUS points – low frequency categories

- The TRAINING data was prepared for you so that certain challenging issues were removed.
- The data were prepared by removing categories that are VERY LOW frequency.
- Low frequency categories cause problems for resampling methods!

BONUS points – low frequency categories

- However, if you would like learn how to deal low frequency categories you may work with the **BONUS** data set supplied on Canvas.
- The **BONUS** data set is larger than the TRAINING set and consists of all categories for the categorical inputs.
- Some of the categories are VERY LOW frequency!

BONUS points – low frequency categories

- You only need to focus on the classification task with the **BONUS** data.
- **You may earn up to 10 BONUS points** if you fit 2 classification models which account for:
 - Low frequency categorical input classes via LUMPING
 - Near zero variance features.
- You **MUST** use `recipes` with `caret` or `tidymodels` directly to earn the **BONUS**.
- The **BONUS** data has an additional variable, `challenge_outcome`, but you do **NOT** need to consider `challenge_outcome` for the low frequency categories BONUS.

BONUS points – imbalanced data

- **IMPORTANT**: As you SHOULD see in your data exploration, the binary outcome is imbalanced.
- However, the binary outcome does not qualify as a “RARE” event since the class imbalance is **NOT** less than 15%.
- However, if you would like learn how to deal with imbalanced data, you may work with the **BONUS** data set provided in Canvas.

BONUS points – imbalanced data

- The **BONUS** data set includes an additional column compared to the TRAINING data set.
- The **BONUS** data set column `challenge_outcome` is a BINARY outcome where $\text{challenge_outcome} = 1$ is the EVENT and $\text{challenge_outcome} = \bar{0}$ is the NON-EVENT.
- The `challenge_outcome` BINARY variable is more difficult because the EVENT $\bar{1}$ is RARE!!
 - The Paint popularity was defined another way...which makes this problem more difficult!

BONUS points – imbalanced data

- You only need to focus on the classification task with the **BONUS** data.
- **You may earn up to 15 BONUS points** if you fit 3 classification models which account for:
 - Low frequency categorical input classes via lumping
 - Near zero variance features
 - Output class imbalance via SUBSAMPLING methods
- You **MUST** use `recipes` with `caret` or `tidymodels` directly to earn the **BONUS**.
- Subsampling is a technique which artificially balances the output classes. This artificial sampling must be accounted for and should NEVER be done manually.
 - Please see [Ch 11 from the caret documentation](#) if you would like to learn about subsampling in `caret`.
 - Please see the [tidymodels Learn page on subsampling](#) for dealing with class imbalance in `tidymodels`.
 - Please see this [Julia Silge blogpost](#) for another `tidymodels` example with class imbalance.

BONUS points – imbalanced data

- The imbalanced data bonus shares some aspects with the low frequency category bonus.
- It is recommended to try the low frequency category bonus first before attempting the imbalanced data bonus.